FIRST Name: __________ LAST Name: ____________ SID (All Digits): __________

- **(5 Points)** On *every* page, print legibly your name and ALL digits of your SID. For every page on which you do not write your name and SID, you forfeit a point, up to the maximum 5 points.
- **(10 Points) (Pledge of Academic Integrity)** Hand-copy, sign, and date the single-line text (which begins with *I have read, ...*) of the Pledge of Academic Integrity on page 3 of this document. Your solutions will *not* be evaluated without this.
- **Urgent Contact with the Teaching Staff:** In case of an urgent matter, raise your hand.
- **This document consists of pages numbered 1 through 11.** Verify that your copy of the exam is free of anomalies, and contains all of the specified number of pages. If you find a defect in your copy, contact the teaching staff immediately.
- This exam is designed to be completed within 70 minutes. However, you may use up to 80 minutes total—*in one sitting*—to tackle it.

  The exam starts at 8:10 pm California time. Your allotted window begins with respect to this start time. Students who have official accommodations of 1.5× and 2× time windows have 120 and 160 minutes, respectively.
- **This exam is closed book.** You may not use or access, or cause to be used or accessed, any reference in print or electronic form at any time during the exam, except one double-sided 8.5" × 11" sheet of handwritten, original notes having no appendage.

  Collaboration is <u>not</u> permitted.

  Computing, communication, and other electronic devices (except dedicated timekeepers) must be turned off.

  Scratch paper will be provided to you; ask for more if you run out. You may not use your own scratch paper.
- Please write neatly and legibly, because *if we can't read it, we can't evaluate it.*
- For each problem, limit your work to the space provided specifically for that problem. *No other work will be considered. For example, we will not evaluate scratch work. No exceptions.*
- Unless explicitly waived by the specific wording of a problem, you must explain your responses (and reasoning) succinctly, but clearly and convincingly.
- In some parts of a problem, we may ask you to establish a certain result—for example, "show this" or "prove that." Even if you're unable to establish the result that we ask of you, you may still take that result for granted—and use it in any subsequent part of the problem.

- If we ask you to provide a "reasonably simple expression" for something, by default we expect your expression to be in closed form—one *not* involving a sum $\sum$ or an integral $\int$—*unless* we explicitly tell you otherwise.
- Noncompliance with these or other instructions from the teaching staff—*including, for example, commencing work prematurely, or continuing it beyond the allocated time window*—is a serious violation of the Code of Student Conduct.

Pledge of Academic Integrity

By my honor, I affirm that

(1) this document—which I have produced for the evaluation of my performance—reflects my original, bona fide work, and that I have neither provided to, nor received from, anyone excessive or unreasonable assistance that produces unfair advantage for me or for any of my peers;

(2) as a member of the UC Berkeley community, I have acted with honesty, integrity, respect for others, and professional responsibility—and in a manner consistent with the letter and intent of the campus Code of Student Conduct;

(3) I have not violated—nor aided or abetted anyone else to violate—the instructions for this exam given by the course staff, including, but not limited to, those on the cover page of this document; and

(4) More generally, I have not committed any act that violates—nor aided or abetted anyone else to violate—UC Berkeley, state, or Federal regulations, during this exam.

(10 Points) In the space below, hand-write the following sentence, verbatim. Then write your name in legible letters, sign, include your full SID, and date before submitting your work:

I have read, I understand, and I commit to adhere to the letter and spirit of the pledge above.

____

____

Full Name: ________________________ Signature: ________________________

Date: ________________________ Student ID: ________________________

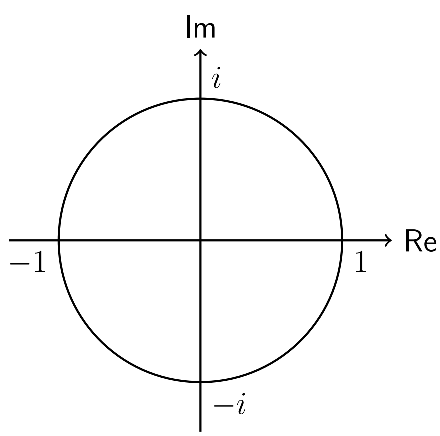
E1.1 (68 Points Points) Discrete-Time (DT) Complex-Exponential Signals

Consider the complex number $z = e^{i\pi/4}$, and construct the DT signal $x : \mathbb{Z} \to \mathbb{C}$ as follows:

$$\forall n \in \mathbb{Z}, \qquad x[n] = z^n = e^{i\pi n/4}.$$

(a) (8 Points) Determine α and β in $z = \alpha + i\beta$, the Cartesian representation of z.

(b) (24 Points) Determine all the eight *distinct* values that the signal x takes, starting from time $n = 0$. Be sure to identify each value and point clearly on the diagram below. You must indicate which value and point on the diagram corresponds to which time-instance value of the signal.



(c) (10 Points) Explain why $\sum_{n=0}^{7} x[n] = 0$. Use a geometric or an algebraic reasoning.

For the remainder of the problem, consider the Cartesian vector $\boldsymbol{x} \in \mathbb{C}^8$ that is constructed from the signal x as follows:

$$\boldsymbol{x} = \begin{bmatrix} x[0] \\ \vdots \\ x[7] \end{bmatrix},$$

and let $\mathbf{1}$ denote the all-ones vector in $\mathbb{C}^8$. In your handwritten solutions, please use the double-struck $\mathbb{1}$ to denote the all-ones vector, and to distinguish it from the scalar 1.

(d) (8 Points) Determine the angle θ that the vector $\boldsymbol{x}$ makes with the all-ones vector. Recall that

$$\cos\theta = \frac{\langle \boldsymbol{x}, \mathbf{1} \rangle}{\|\boldsymbol{x}\| \, \|\mathbf{1}\|},$$

where the inner product $\langle \boldsymbol{x}, \boldsymbol{y} \rangle \triangleq \boldsymbol{x}^\mathsf{T} \boldsymbol{y}^*$ for any two vectors $\boldsymbol{x}$ and $\boldsymbol{y}$ in $\mathbb{C}^n$.

(e) (18 Points) Determine each of the following norms for the signal x (or, equivalently, the vector $\boldsymbol{x}$).

(i) (6 Points) $\|x\| = \|\boldsymbol{x}\|$ (i.e., the 2-norm, also known as the Euclidean norm).

(ii) (6 Points) $\|x\|_1 = \|\boldsymbol{x}\|_1$.

(iii) (6 Points) $\|x\|_\infty = \|\boldsymbol{x}\|_\infty$.

E1.2 (67 Points Points) On Finite-Energy Discrete-Time (DT) Signals

Let $\mathcal{V} = [\mathbb{Z} \to \mathbb{R}]$ denote the vector space of all real-valued discrete-time (DT) signals.

We define the inner product of any two signals f and g in $\mathcal{V}$ as

$$\langle f, g \rangle \triangleq \sum_{-\infty}^{\infty} f[n]\, g[n].$$

Thus, $\mathcal{V}$ is an *inner product space*—also known as a *Euclidean space*—of DT signals.

Now consider a subset $\mathcal{E}$ of $\mathcal{V}$ consisting of all *finite-energy* DT signals. That is,

$$\mathcal{E} = \left\{ x : \mathbb{Z} \to \mathbb{R} \,\middle|\, E_x < \infty \right\},$$

where the energy of the signal f is

$$E_x \triangleq \|x\|^2 = \langle x, x \rangle = \sum_{n=-\infty}^{\infty} |x[n]|^2,$$

and $\| \cdot \|$ denotes the Euclidean (ℓ_2) norm. Finite-energy signals are sometimes called ℓ_2 *signals* or *energy signals*. And the set $\mathcal{E}$ is often called $\ell_2(\mathbb{Z})$.

Throughout this problem, let x, y, and z denote finite-energy DT signals (i.e., ℓ_2 signals).

(a) (10 Points) Consider a DT signal z whose energy $E_z = 0$. Determine z completely, and explain why it must be unique.

(b) (25 Points) Provide a succinct, yet clear and convincing explanation as to why $\mathcal{E}$ is a subspace. To do this you must establish the following:

(i) (5 Points) Existence of a Zero Signal (You need *not* repeat work that you've already done.)

(ii) (10 Points) Closure Under Scalar Multiplication of a Signal

(iii) (10 Points) Closure Under Signal Addition

(c) (16 Points) True or False?

$$\|x + y\| = \|x - y\| \text{ if, and only if, } x \perp y.$$

If you assert the claim to be true, prove it. If you assert that it's false, find a pair of signals x and y in $\mathcal{E}$ and show that they provide a counterexample.

(d) (16 Points) Prove that $\langle x + y, x - y \rangle = 0$ if, and only if, $\|x\| = \|y\|$.

E1.3 (50 Points Points) On the Geometry of Continuous-Time (CT) Signal Spaces

Consider the vector space $\mathcal{S}$ of real-valued CT signals continuous on the interval $[-1, +1]$.

We define the following inner product in the signal space $\mathcal{S}$:

$$\forall f, g \in \mathcal{S}, \qquad \langle f, g \rangle \triangleq \int_{-1}^{+1} f(t) g(t)\, dt.$$

The Euclidean norm induced by the inner product is $\|f\| \triangleq \sqrt{\langle f, f \rangle}$, and the angle θ_{fg} between any two signals f and g in $\mathcal{S}$ satisfies the equation

$$\cos \theta_{fg} = \frac{\langle f, g \rangle}{\|f\|\, \|g\|}.$$

Consider the following three signals x, y, and z in $\mathcal{S}$:

$$\forall t \in \mathbb{R}, \qquad x(t) = \frac{1}{\sqrt{2}}, \qquad y(t) = \frac{\sqrt{3}}{\sqrt{2}}\, t, \qquad \text{and} \qquad z(t) = \frac{1}{2} + \frac{\sqrt{3}}{2} t.$$

(a) (20 Points) Show that $\|x\| = \|y\| = \|z\| = 1$.

(b) (12 Points) Determine whether the set of signals $\{x, y, z\}$ is linearly independent. Provide a succinct, yet clear and convincing explanation.

For the remainder of this problem, it may or may not be useful for you to recall that

$$\cos\left(\frac{\pi}{4}\right) = \sin\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}.$$

(c) (18 Points) Determine numerically, as a multiple of π (not as a formula), the following angles:

(i) (6 Points) θ_{xy}, the angle between signals x and y.

(ii) (6 Points) θ_{xz}, the angle between signals x and z.

(iii) (6 Points) θ_{yz}, the angle between signals y and z.